

# Agenda

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Hypothesis & significance testing: short review

Tests for a population mean (normal random sample with a small sample size)

Tests for a population proportion

Tests for two populations

# Tests about a population mean of a normal r.v. with known variance (Hypothesis Testing)

$$H_0: \mu = \mu_0$$

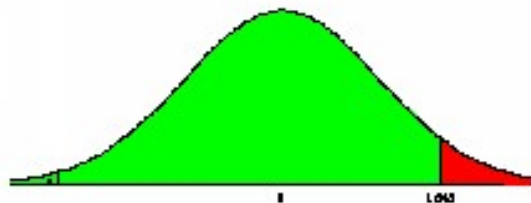
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$$H_1: \mu > \mu_0$$

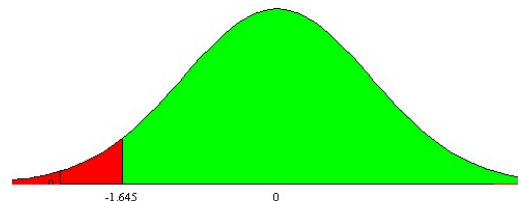
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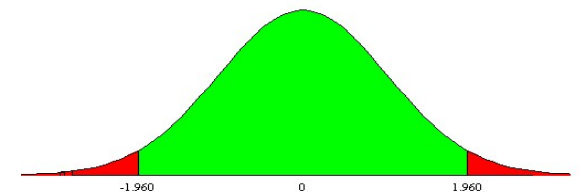
$$\text{R.R. : } z \geq z_\alpha$$

$$H_1: \mu < \mu_0$$



$$\text{R.R. : } z \leq -z_\alpha$$

$$H_1: \mu \neq \mu_0$$



$$\begin{aligned} \text{R.R. : } z &\leq -z_{\alpha/2} \\ \text{or } z &\geq z_{\alpha/2} \end{aligned}$$

# Tests about a population mean of a r.v. with large sample size (Hypothesis Testing)

$$H_0: \mu = \mu_0$$

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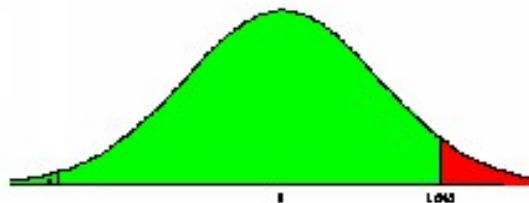
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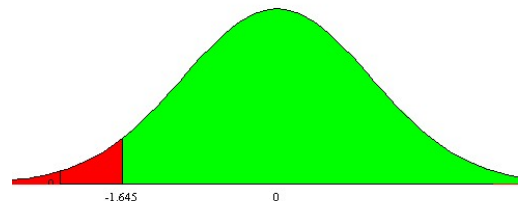
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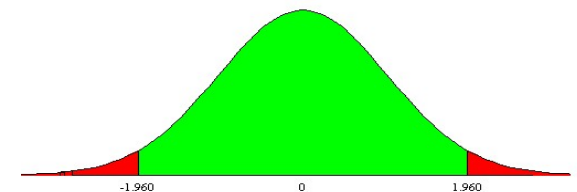
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# Test procedure

Hypothesis testing:

$\alpha \rightarrow$  Rejection Region  $\rightarrow$  conclusion  
computation of the comparison of  
critical value,  $k$  ( $\bar{x}$  and  $k$ ) or ( $z$  and  $z_\alpha$ )

Significance testing:

$\bar{x} \rightarrow$   $P\{\text{type I error when } H_0 \text{ is rejected with } k = \bar{x}\}$   $\rightarrow$  conclusion  
computation of the comparison of  
P-value,  $P$  P-value and  $\alpha$

Why? Preference on  $\alpha$  may change,  $P$ -value gives more information.

# Tests about a population mean of a normal r.v. with known variance (Significance Testing)

$$H_0: \mu = \mu_0$$

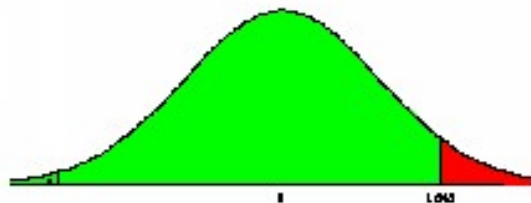
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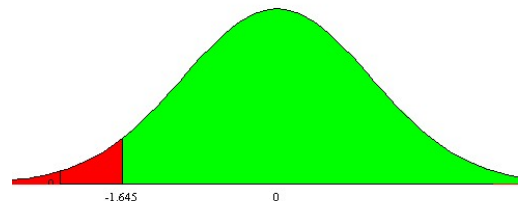
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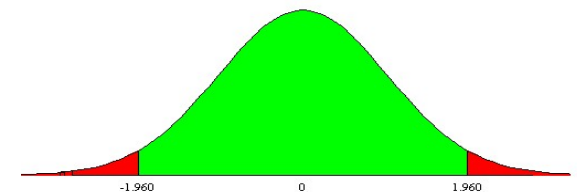
P-value:  $P(Z \geq z_0)$

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P-value:  $P(Z \leq z_0)$

$$H_1: \mu \neq \mu_0$$



P-value:  $2 \times P(Z \geq |z_0|)$

# Tests about a population mean of a r.v. with large sample size (Significance Testing)

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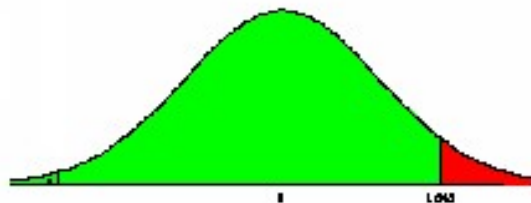
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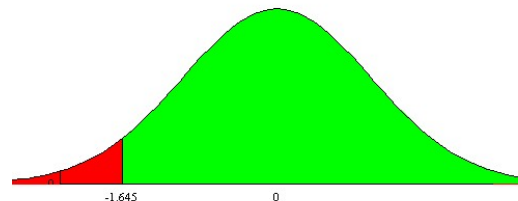
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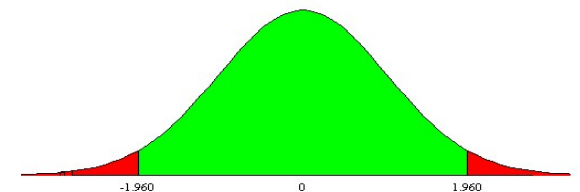
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# P-values

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Definition: The p-value (or observed significance level) is the smallest level of significance at which  $H_0$  would be rejected when a specified test procedure is used on a given data set. Once the p-value has been determined, the conclusion at any particular level  $\alpha$  results from comparing the P-value to  $\alpha$ :

- 1)  $P\text{-value} \leq \alpha \rightarrow \text{reject } H_0 \text{ at level } \alpha$
- 2)  $P\text{-value} > \alpha \rightarrow \text{cannot reject } H_0 \text{ at level } \alpha$

# Tests about a population mean of a normal r.v. with unknown variance and small sample size

$$H_0: \mu = \mu_0 \quad \text{Test statistics: } T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}$$

Test statistics is given under the assumption that  $H_0$  is true.

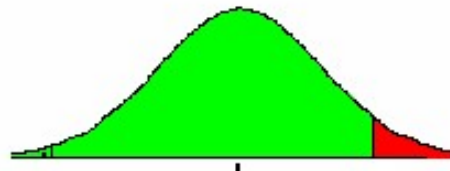
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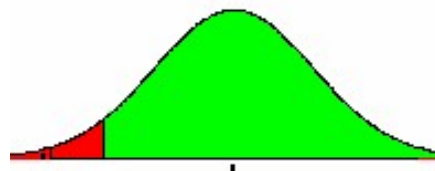
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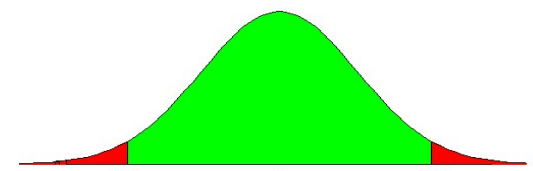
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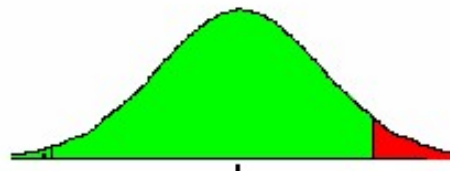
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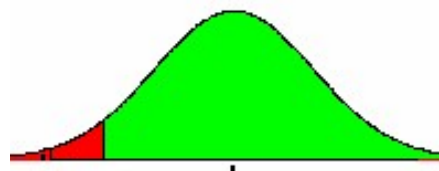
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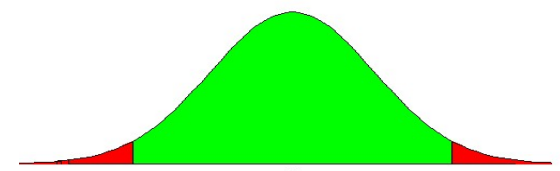
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P-value:  $P(T \geq t_0)$



P-value:  $P(T \leq t_0)$



P-value:  $2 \times P(T \geq |t_0|)$

**Find bounds on the p-value using the t-table.**

$$\text{Test statistic: } \frac{\bar{X} - \mu_0}{s/\sqrt{n}} = \frac{111.38 - 100}{19.28/\sqrt{15}} = 2.286$$

## Example



Service times of incoming calls in an airline reservation system:  $S_i \sim N(\mu, \sigma)$  with  $\sigma$  is unknown.

Average service time in the biggest 10 airline companies is reported as 100 seconds.

$$H_0: \mu = 100$$

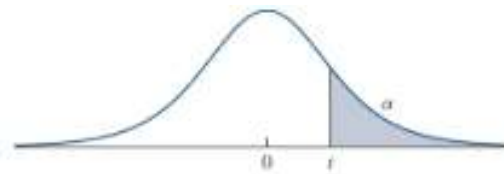
If the average is 100 or less, there will be no change in the system.

$$H_a: \mu > 100$$

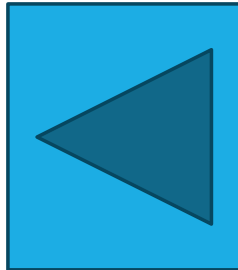
If the average has increased above 100, more capacity is needed, i.e., number of servers in the call center should be increased.

$$\bar{x} = 111.38 \quad n = 15 \quad \alpha = 0.05 \quad s = 19.28$$

Is more capacity needed?



| v        | $\alpha$ |       |       |       |        |        |        |         |         |
|----------|----------|-------|-------|-------|--------|--------|--------|---------|---------|
|          | 0.40     | 0.25  | 0.10  | 0.05  | 0.025  | 0.01   | 0.005  | 0.001   | 0.0005  |
| 1        | 0.325    | 1.000 | 3.078 | 6.314 | 12.706 | 31.821 | 63.657 | 318.309 | 636.619 |
| 2        | 0.289    | 0.816 | 1.886 | 2.920 | 4.303  | 6.965  | 9.925  | 22.327  | 31.599  |
| 3        | 0.277    | 0.765 | 1.638 | 2.353 | 3.182  | 4.541  | 5.841  | 10.215  | 12.924  |
| 4        | 0.271    | 0.741 | 1.533 | 2.132 | 2.776  | 3.747  | 4.604  | 7.173   | 8.610   |
| 5        | 0.267    | 0.727 | 1.476 | 2.015 | 2.571  | 3.365  | 4.032  | 5.893   | 6.869   |
| 6        | 0.265    | 0.718 | 1.440 | 1.943 | 2.447  | 3.143  | 3.707  | 5.208   | 5.959   |
| 7        | 0.263    | 0.711 | 1.415 | 1.895 | 2.365  | 2.998  | 3.499  | 4.785   | 5.408   |
| 8        | 0.262    | 0.706 | 1.397 | 1.860 | 2.306  | 2.896  | 3.355  | 4.501   | 5.041   |
| 9        | 0.261    | 0.703 | 1.383 | 1.833 | 2.262  | 2.821  | 3.250  | 4.297   | 4.781   |
| 10       | 0.260    | 0.700 | 1.372 | 1.812 | 2.228  | 2.764  | 3.169  | 4.144   | 4.587   |
| 11       | 0.260    | 0.697 | 1.363 | 1.796 | 2.201  | 2.718  | 3.106  | 4.025   | 4.437   |
| 12       | 0.259    | 0.695 | 1.356 | 1.782 | 2.179  | 2.681  | 3.055  | 3.930   | 4.318   |
| 13       | 0.259    | 0.694 | 1.350 | 1.771 | 2.160  | 2.650  | 3.012  | 3.852   | 4.221   |
| 14       | 0.258    | 0.692 | 1.345 | 1.761 | 2.145  | 2.624  | 2.977  | 3.787   | 4.140   |
| 15       | 0.258    | 0.691 | 1.341 | 1.753 | 2.131  | 2.602  | 2.947  | 3.733   | 4.073   |
| 16       | 0.258    | 0.690 | 1.337 | 1.746 | 2.120  | 2.583  | 2.921  | 3.686   | 4.015   |
| 17       | 0.257    | 0.689 | 1.333 | 1.740 | 2.110  | 2.567  | 2.898  | 3.646   | 3.965   |
| 18       | 0.257    | 0.688 | 1.330 | 1.734 | 2.101  | 2.552  | 2.878  | 3.610   | 3.922   |
| 19       | 0.257    | 0.688 | 1.328 | 1.729 | 2.093  | 2.539  | 2.861  | 3.579   | 3.883   |
| 20       | 0.257    | 0.687 | 1.325 | 1.725 | 2.086  | 2.528  | 2.845  | 3.552   | 3.850   |
| 21       | 0.257    | 0.686 | 1.323 | 1.721 | 2.080  | 2.518  | 2.831  | 3.527   | 3.819   |
| 22       | 0.256    | 0.686 | 1.321 | 1.717 | 2.074  | 2.508  | 2.819  | 3.505   | 3.792   |
| 23       | 0.256    | 0.685 | 1.319 | 1.714 | 2.069  | 2.500  | 2.807  | 3.485   | 3.768   |
| 24       | 0.256    | 0.685 | 1.318 | 1.711 | 2.064  | 2.492  | 2.797  | 3.467   | 3.745   |
| 25       | 0.256    | 0.684 | 1.316 | 1.708 | 2.060  | 2.485  | 2.787  | 3.450   | 3.725   |
| 26       | 0.256    | 0.684 | 1.315 | 1.706 | 2.056  | 2.479  | 2.779  | 3.435   | 3.707   |
| 27       | 0.256    | 0.684 | 1.314 | 1.703 | 2.052  | 2.473  | 2.771  | 3.421   | 3.690   |
| 28       | 0.256    | 0.683 | 1.313 | 1.701 | 2.048  | 2.467  | 2.763  | 3.408   | 3.674   |
| 29       | 0.256    | 0.683 | 1.311 | 1.699 | 2.045  | 2.462  | 2.756  | 3.396   | 3.659   |
| 30       | 0.256    | 0.683 | 1.310 | 1.697 | 2.042  | 2.457  | 2.750  | 3.385   | 3.646   |
| 35       | 0.255    | 0.682 | 1.306 | 1.690 | 2.030  | 2.438  | 2.724  | 3.340   | 3.591   |
| 40       | 0.255    | 0.681 | 1.303 | 1.684 | 2.021  | 2.423  | 2.704  | 3.307   | 3.551   |
| 60       | 0.254    | 0.679 | 1.296 | 1.671 | 2.000  | 2.390  | 2.660  | 3.232   | 3.460   |
| 120      | 0.254    | 0.677 | 1.289 | 1.658 | 1.980  | 2.358  | 2.617  | 3.160   | 3.373   |
| $\infty$ | 0.253    | 0.674 | 1.282 | 1.645 | 1.960  | 2.326  | 2.576  | 3.090   | 3.291   |



# Example

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$$H_0: \mu = 100$$

$$H_a: \mu > 100$$

$$\bar{x} = 111.38$$

$$s = 19.28$$

$$n = 15$$

Find bounds on the p-value. Is more capacity needed at  $\alpha=0.01$ ?  $\alpha=0.05$ ?

$$0.01 < \text{p-value} < 0.025$$

# Large-sample tests for a population proportion

- $X$  : # of successes in a random sample with  $n \ll$  population size
- $p$  : proportion of 'success' in a population
- $X \sim \text{bin}(n, p)$ 
  - For large  $n$  with  $np \geq 10$  and  $n(1-p) \geq 10$ ,  $X$  is approximately normal with  $E(X) = np$ , &  $V(X) = np(1-p)$ .
  - $\hat{p} = X/n$  is unbiased since  $E(\hat{p}) = p$ , has approximately a normal distribution, and its standard deviation is  $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$ .

$$H_0: p = p_0$$

true only if  
 $np_0 \geq 10$   
&  $n(1-p_0) \geq 10$

When  $H_0$  is true, 
$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}} \sim N(0,1)$$

# Test statistics, $H_a$ & RR for population proportion $p$

$$H_0: p = p_0 \quad \text{Test statistics: } Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

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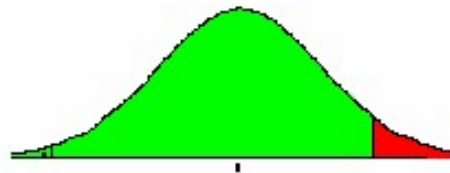
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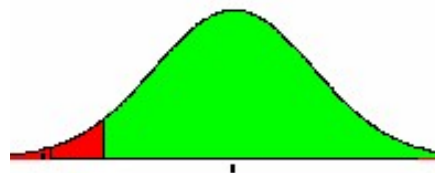
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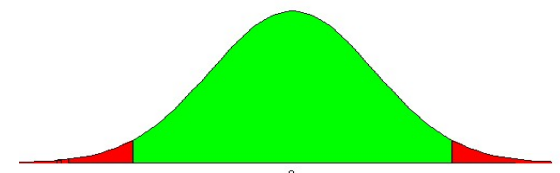
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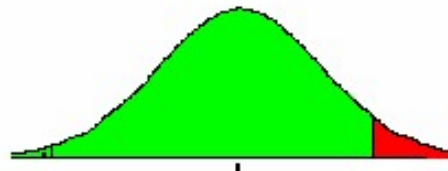
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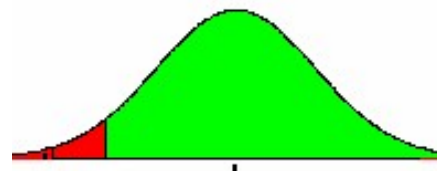
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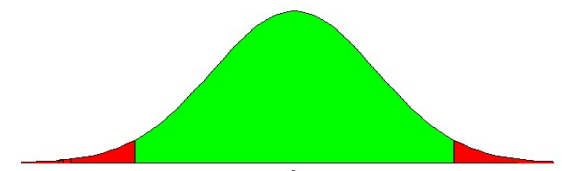
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P-value:  $P(Z \leq z_0)$



P-value:  $2 \times P(Z \geq |z_0|)$

## Power and sample size in a t-test

- Complicated → You can find appropriate functions in Python, but we will not use them.

# Tests about means of two normal populations with known variance

$$H_0: \mu_1 - \mu_2 = \Delta_0$$

$$\text{Test statistics: } Z = \frac{\bar{X} - \bar{Y} - \Delta_0}{\sqrt{\frac{\sigma_1^2}{m} + \frac{\sigma_2^2}{n}}}$$

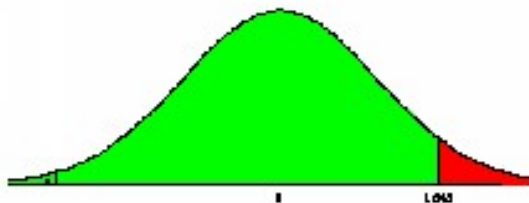
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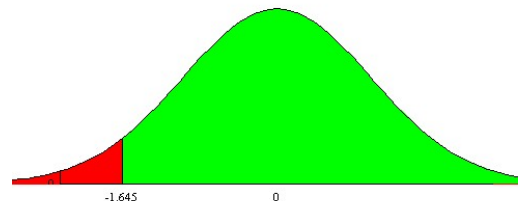
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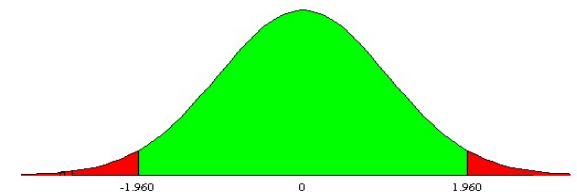
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# Tests about means of two populations with large sample size (unknown variance)

$$H_0: \mu_1 - \mu_2 = \Delta_0$$

Test statistics:

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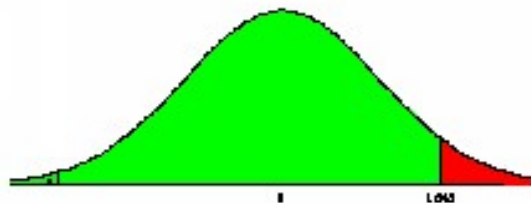
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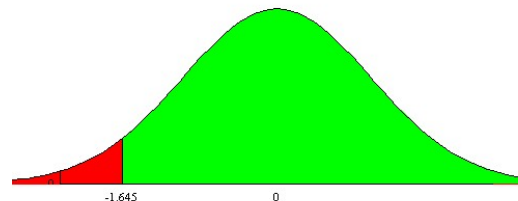
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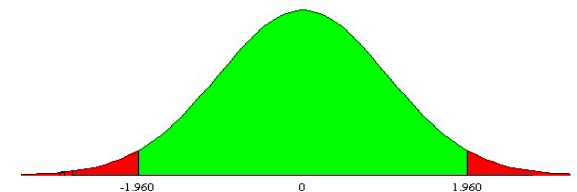
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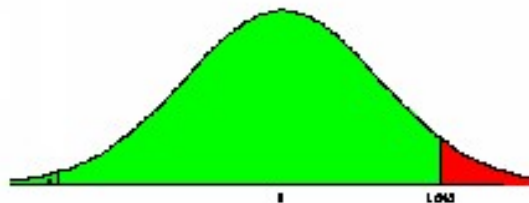
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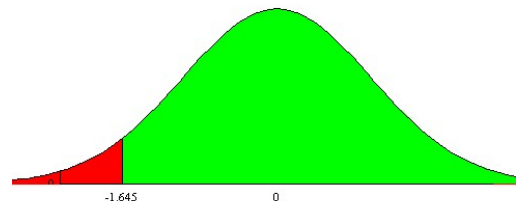
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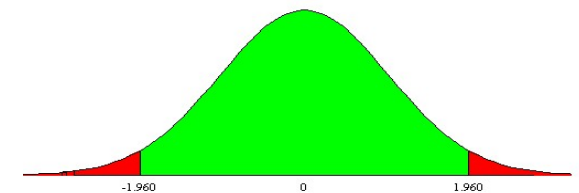
Reject



$$\text{R.R.: } t \geq t_{\alpha, \nu}$$



$$\text{R.R.: } t \leq -t_{\alpha, \nu}$$



$$\begin{aligned} \text{R.R.: } & t \leq -t_{\alpha/2, \nu} \\ \text{or } & t \geq t_{\alpha/2, \nu} \end{aligned}$$

# Class Exercise

---

The production manager wants to compare the average weights of popcorns from two production lines. Experience shows that the weights of popcorns can be assumed to be normal. He would like to know whether the true mean of the weights of popcorns from two production lines are different than each other. He obtained the following data:

|   | Distribution of the weights | Sample size ( $n_1, n_2$ ) | True standard deviation known/unknown ( $(\sigma_1, \sigma_2) / (s_1, s_2)$ ) |
|---|-----------------------------|----------------------------|-------------------------------------------------------------------------------|
| 1 | Normal                      | (12, 16)                   | $(\sigma_1, \sigma_2) = (0.5, 0.6)$                                           |
| 2 | Unknown                     | (12, 16)                   | $(s_1, s_2) = (0.5, 0.6)$                                                     |
| 3 | Normal                      | (12, 16)                   | $(s_1, s_2) = (0.5, 0.6)$                                                     |
| 4 | Unknown                     | (42, 46)                   | $(s_1, s_2) = (0.5, 0.6)$                                                     |

Note that the corresponding degrees of freedom is 25 for Case 3.

## Class Exercise – (1)

---

Distribution of the weights: Normal

Sample size ( $n_1, n_2$ ): (12, 16)

$(\sigma_1, \sigma_2) = (0.5, 0.6)$

$$\text{Test statistic} = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}} = \frac{5.2 - 5.61}{\sqrt{\frac{0.5^2}{12} + \frac{0.6^2}{16}}} = -1.9696$$

## Class Exercise – (1)

---

Distribution of the weights: Normal

Sample size ( $n_1, n_2$ ): (12, 16)

$(\sigma_1, \sigma_2) = (0.5, 0.6)$

$$\text{Test statistic} = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}} = \frac{5.2 - 5.6}{\sqrt{\frac{0.5^2}{12} + \frac{0.6^2}{16}}} = -1.92$$

Test Statistics:

$$z_0 = -1.9696$$

P-value:

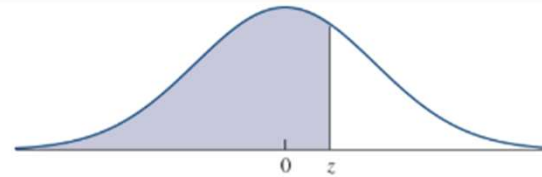
$$2 \times P(Z \geq |z_0|)$$

The p-value:

$$2 \times P(Z > |-1.97|) = 2(1 - 0.9756) \\ = 2 \times 0.0244 = 0.0488 < 0.05$$

→ Reject  $H_0$  at  $\alpha=5\%$ .

→ Cannot reject  $H_0$  at  $\alpha=1\%$ .



| z   | 0.00  | 0.01  | 0.02  | 0.03  | 0.04  | 0.05  | 0.06  | 0.07  | 0.08  | 0.09  |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 0.0 | .5000 | .5040 | .5080 | .5120 | .5160 | .5199 | .5239 | .5279 | .5319 | .5359 |
| 0.1 | .5398 | .5438 | .5478 | .5517 | .5557 | .5596 | .5636 | .5675 | .5714 | .5753 |
| 0.2 | .5793 | .5832 | .5871 | .5910 | .5948 | .5987 | .6026 | .6064 | .6103 | .6141 |
| 0.3 | .6179 | .6217 | .6255 | .6293 | .6331 | .6368 | .6406 | .6443 | .6480 | .6517 |
| 0.4 | .6554 | .6591 | .6628 | .6664 | .6700 | .6736 | .6772 | .6808 | .6844 | .6879 |
| 0.5 | .6915 | .6950 | .6985 | .7019 | .7054 | .7088 | .7123 | .7157 | .7190 | .7224 |
| 0.6 | .7257 | .7291 | .7324 | .7357 | .7389 | .7422 | .7454 | .7486 | .7517 | .7549 |
| 0.7 | .7580 | .7611 | .7642 | .7673 | .7704 | .7734 | .7764 | .7794 | .7823 | .7852 |
| 0.8 | .7881 | .7910 | .7939 | .7967 | .7995 | .8023 | .8051 | .8078 | .8106 | .8133 |
| 0.9 | .8159 | .8186 | .8212 | .8238 | .8264 | .8289 | .8315 | .8340 | .8365 | .8389 |
| 1.0 | .8413 | .8438 | .8461 | .8485 | .8508 | .8531 | .8554 | .8577 | .8599 | .8621 |
| 1.1 | .8643 | .8665 | .8686 | .8708 | .8729 | .8749 | .8770 | .8790 | .8810 | .8830 |
| 1.2 | .8849 | .8869 | .8888 | .8907 | .8925 | .8944 | .8962 | .8980 | .8997 | .9015 |
| 1.3 | .9032 | .9049 | .9066 | .9082 | .9099 | .9115 | .9131 | .9147 | .9162 | .9177 |
| 1.4 | .9192 | .9207 | .9222 | .9236 | .9251 | .9265 | .9279 | .9292 | .9306 | .9319 |
| 1.5 | .9332 | .9345 | .9357 | .9370 | .9382 | .9394 | .9406 | .9418 | .9429 | .9441 |
| 1.6 | .9452 | .9463 | .9474 | .9484 | .9495 | .9505 | .9515 | .9525 | .9535 | .9545 |
| 1.7 | .9554 | .9564 | .9573 | .9582 | .9591 | .9599 | .9608 | .9616 | .9625 | .9633 |
| 1.8 | .9641 | .9649 | .9656 | .9664 | .9671 | .9678 | .9686 | .9693 | .9699 | .9706 |
| 1.9 | .9713 | .9719 | .9726 | .9732 | .9738 | .9744 | .9750 | .9756 | .9761 | .9767 |
| 2.0 | .9772 | .9778 | .9783 | .9788 | .9793 | .9798 | .9803 | .9808 | .9812 | .9817 |
| 2.1 | .9821 | .9826 | .9830 | .9834 | .9838 | .9842 | .9846 | .9850 | .9854 | .9857 |
| 2.2 | .9861 | .9864 | .9868 | .9871 | .9875 | .9878 | .9881 | .9884 | .9887 | .9890 |
| 2.3 | .9893 | .9896 | .9898 | .9901 | .9904 | .9906 | .9909 | .9911 | .9913 | .9916 |
| 2.4 | .9918 | .9920 | .9922 | .9925 | .9927 | .9929 | .9931 | .9932 | .9934 | .9936 |
| 2.5 | .9938 | .9940 | .9941 | .9943 | .9945 | .9946 | .9948 | .9949 | .9951 | .9952 |
| 2.6 | .9953 | .9955 | .9956 | .9957 | .9959 | .9960 | .9961 | .9962 | .9963 | .9964 |
| 2.7 | .9965 | .9966 | .9967 | .9968 | .9969 | .9970 | .9971 | .9972 | .9973 | .9974 |
| 2.8 | .9974 | .9975 | .9976 | .9977 | .9977 | .9978 | .9979 | .9979 | .9980 | .9981 |
| 2.9 | .9981 | .9982 | .9982 | .9983 | .9984 | .9984 | .9985 | .9985 | .9986 | .9986 |

Test Statistics:

$$z_0 = -1.92$$

P-value:

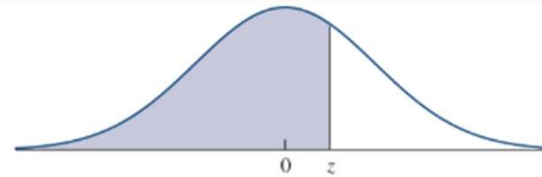
$$2 \times P(Z \geq |z_0|)$$

The p-value:

$$2 \times P(Z > |-1.92|) = 2(1 - 0.9726) \\ = 2 \times 0.0274 = 0.0548 > 0.05$$

→ Cannot reject  $H_0$  at  $\alpha=5\%$ .

→ Cannot reject  $H_0$  at  $\alpha=1\%$ .



| z   | 0.00  | 0.01  | 0.02  | 0.03  | 0.04  | 0.05  | 0.06  | 0.07  | 0.08  | 0.09  |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 0.0 | .5000 | .5040 | .5080 | .5120 | .5160 | .5199 | .5239 | .5279 | .5319 | .5359 |
| 0.1 | .5398 | .5438 | .5478 | .5517 | .5557 | .5596 | .5636 | .5675 | .5714 | .5753 |
| 0.2 | .5793 | .5832 | .5871 | .5910 | .5948 | .5987 | .6026 | .6064 | .6103 | .6141 |
| 0.3 | .6179 | .6217 | .6255 | .6293 | .6331 | .6368 | .6406 | .6443 | .6480 | .6517 |
| 0.4 | .6554 | .6591 | .6628 | .6664 | .6700 | .6736 | .6772 | .6808 | .6844 | .6879 |
| 0.5 | .6915 | .6950 | .6985 | .7019 | .7054 | .7088 | .7123 | .7157 | .7190 | .7224 |
| 0.6 | .7257 | .7291 | .7324 | .7357 | .7389 | .7422 | .7454 | .7486 | .7517 | .7549 |
| 0.7 | .7580 | .7611 | .7642 | .7673 | .7704 | .7734 | .7764 | .7794 | .7823 | .7852 |
| 0.8 | .7881 | .7910 | .7939 | .7967 | .7995 | .8023 | .8051 | .8078 | .8106 | .8133 |
| 0.9 | .8159 | .8186 | .8212 | .8238 | .8264 | .8289 | .8315 | .8340 | .8365 | .8389 |
| 1.0 | .8413 | .8438 | .8461 | .8485 | .8508 | .8531 | .8554 | .8577 | .8599 | .8621 |
| 1.1 | .8643 | .8665 | .8686 | .8708 | .8729 | .8749 | .8770 | .8790 | .8810 | .8830 |
| 1.2 | .8849 | .8869 | .8888 | .8907 | .8925 | .8944 | .8962 | .8980 | .8997 | .9015 |
| 1.3 | .9032 | .9049 | .9066 | .9082 | .9099 | .9115 | .9131 | .9147 | .9162 | .9177 |
| 1.4 | .9192 | .9207 | .9222 | .9236 | .9251 | .9265 | .9279 | .9292 | .9306 | .9319 |
| 1.5 | .9332 | .9345 | .9357 | .9370 | .9382 | .9394 | .9406 | .9418 | .9429 | .9441 |
| 1.6 | .9452 | .9463 | .9474 | .9484 | .9495 | .9505 | .9515 | .9525 | .9535 | .9545 |
| 1.7 | .9554 | .9564 | .9573 | .9582 | .9591 | .9599 | .9608 | .9616 | .9625 | .9633 |
| 1.8 | .9641 | .9649 | .9656 | .9664 | .9671 | .9678 | .9686 | .9693 | .9699 | .9706 |
| 1.9 | .9713 | .9719 | .9726 | .9732 | .9738 | .9744 | .9750 | .9756 | .9761 | .9767 |
| 2.0 | .9772 | .9778 | .9783 | .9788 | .9793 | .9798 | .9803 | .9808 | .9812 | .9817 |
| 2.1 | .9821 | .9826 | .9830 | .9834 | .9838 | .9842 | .9846 | .9850 | .9854 | .9857 |
| 2.2 | .9861 | .9864 | .9868 | .9871 | .9875 | .9878 | .9881 | .9884 | .9887 | .9890 |
| 2.3 | .9893 | .9896 | .9898 | .9901 | .9904 | .9906 | .9909 | .9911 | .9913 | .9916 |
| 2.4 | .9918 | .9920 | .9922 | .9925 | .9927 | .9929 | .9931 | .9932 | .9934 | .9936 |
| 2.5 | .9938 | .9940 | .9941 | .9943 | .9945 | .9946 | .9948 | .9949 | .9951 | .9952 |
| 2.6 | .9953 | .9955 | .9956 | .9957 | .9959 | .9960 | .9961 | .9962 | .9963 | .9964 |
| 2.7 | .9965 | .9966 | .9967 | .9968 | .9969 | .9970 | .9971 | .9972 | .9973 | .9974 |
| 2.8 | .9974 | .9975 | .9976 | .9977 | .9977 | .9978 | .9979 | .9979 | .9980 | .9981 |
| 2.9 | .9981 | .9982 | .9982 | .9983 | .9984 | .9984 | .9985 | .9985 | .9986 | .9986 |

## Class Exercise – (2)

---

Distribution of the weights: Unknown

Sample size ( $n_1, n_2$ ): (12, 16)

$(s_1, s_2) = (0.5, 0.6)$

Cannot carry out the test since we do not know the distribution of the test statistic.

## Class Exercise – (3)

---

Distribution of the weights: Normal

Sample size ( $n_1, n_2$ ): (12, 16)

( $s_1, s_2$ ) = (0.5, 0.6)

Test statistic=

$$\frac{\bar{X} - \bar{Y}}{\sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}} = \frac{5.2 - 5.61}{\sqrt{\frac{0.5^2}{12} + \frac{0.6^2}{16}}} = -1.9696$$

## Class Exercise – (3)

---

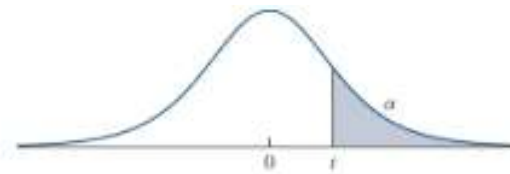
Distribution of the weights: Normal

Sample size ( $n_1, n_2$ ): (12, 16)

( $s_1, s_2$ ) = (0.5, 0.6)

Test statistic=

$$\frac{\bar{X} - \bar{Y}}{\sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}} = \frac{5.2 - 5.6}{\sqrt{\frac{0.5^2}{12} + \frac{0.6^2}{16}}} = -1.9696$$



|          | $\alpha$ |       |       |       |        |        |        |         |         |  |
|----------|----------|-------|-------|-------|--------|--------|--------|---------|---------|--|
| $\nu$    | 0.40     | 0.25  | 0.10  | 0.05  | 0.025  | 0.01   | 0.005  | 0.001   | 0.0005  |  |
| 1        | 0.325    | 1.000 | 3.078 | 6.314 | 12.706 | 31.821 | 63.657 | 318.309 | 636.619 |  |
| 2        | 0.289    | 0.816 | 1.886 | 2.920 | 4.303  | 6.965  | 9.925  | 22.327  | 31.599  |  |
| 3        | 0.277    | 0.765 | 1.638 | 2.353 | 3.182  | 4.541  | 5.841  | 10.215  | 12.924  |  |
| 4        | 0.271    | 0.741 | 1.533 | 2.132 | 2.776  | 3.747  | 4.604  | 7.173   | 8.610   |  |
| 5        | 0.267    | 0.727 | 1.476 | 2.015 | 2.571  | 3.365  | 4.032  | 5.893   | 6.869   |  |
| 6        | 0.265    | 0.718 | 1.440 | 1.943 | 2.447  | 3.143  | 3.707  | 5.208   | 5.959   |  |
| 7        | 0.263    | 0.711 | 1.415 | 1.895 | 2.365  | 2.998  | 3.499  | 4.785   | 5.408   |  |
| 8        | 0.262    | 0.706 | 1.397 | 1.860 | 2.306  | 2.896  | 3.355  | 4.501   | 5.041   |  |
| 9        | 0.261    | 0.703 | 1.383 | 1.833 | 2.262  | 2.821  | 3.250  | 4.297   | 4.781   |  |
| 10       | 0.260    | 0.700 | 1.372 | 1.812 | 2.228  | 2.764  | 3.169  | 4.144   | 4.587   |  |
| 11       | 0.260    | 0.697 | 1.363 | 1.796 | 2.201  | 2.718  | 3.106  | 4.025   | 4.437   |  |
| 12       | 0.259    | 0.695 | 1.356 | 1.782 | 2.179  | 2.681  | 3.055  | 3.930   | 4.318   |  |
| 13       | 0.259    | 0.694 | 1.350 | 1.771 | 2.160  | 2.650  | 3.012  | 3.852   | 4.221   |  |
| 14       | 0.258    | 0.692 | 1.345 | 1.761 | 2.145  | 2.624  | 2.977  | 3.787   | 4.140   |  |
| 15       | 0.258    | 0.691 | 1.341 | 1.753 | 2.131  | 2.602  | 2.947  | 3.733   | 4.073   |  |
| 16       | 0.258    | 0.690 | 1.337 | 1.746 | 2.120  | 2.583  | 2.921  | 3.686   | 4.015   |  |
| 17       | 0.257    | 0.689 | 1.333 | 1.740 | 2.110  | 2.567  | 2.898  | 3.646   | 3.965   |  |
| 18       | 0.257    | 0.688 | 1.330 | 1.734 | 2.101  | 2.552  | 2.878  | 3.610   | 3.922   |  |
| 19       | 0.257    | 0.688 | 1.328 | 1.729 | 2.093  | 2.539  | 2.861  | 3.579   | 3.883   |  |
| 20       | 0.257    | 0.687 | 1.325 | 1.725 | 2.086  | 2.528  | 2.845  | 3.552   | 3.850   |  |
| 21       | 0.257    | 0.686 | 1.323 | 1.721 | 2.080  | 2.518  | 2.831  | 3.527   | 3.819   |  |
| 22       | 0.256    | 0.686 | 1.321 | 1.717 | 2.074  | 2.508  | 2.819  | 3.505   | 3.792   |  |
| 23       | 0.256    | 0.685 | 1.319 | 1.714 | 2.069  | 2.500  | 2.807  | 3.485   | 3.768   |  |
| 24       | 0.256    | 0.685 | 1.318 | 1.711 | 2.064  | 2.492  | 2.797  | 3.467   | 3.745   |  |
| 25       | 0.256    | 0.684 | 1.316 | 1.708 | 2.060  | 2.485  | 2.787  | 3.450   | 3.725   |  |
| 26       | 0.256    | 0.684 | 1.315 | 1.706 | 2.056  | 2.479  | 2.779  | 3.435   | 3.707   |  |
| 27       | 0.256    | 0.684 | 1.314 | 1.703 | 2.052  | 2.473  | 2.771  | 3.421   | 3.690   |  |
| 28       | 0.256    | 0.683 | 1.313 | 1.701 | 2.048  | 2.467  | 2.763  | 3.408   | 3.674   |  |
| 29       | 0.256    | 0.683 | 1.311 | 1.699 | 2.045  | 2.462  | 2.756  | 3.396   | 3.659   |  |
| 30       | 0.256    | 0.683 | 1.310 | 1.697 | 2.042  | 2.457  | 2.750  | 3.385   | 3.646   |  |
| 35       | 0.255    | 0.682 | 1.306 | 1.690 | 2.030  | 2.438  | 2.724  | 3.340   | 3.591   |  |
| 40       | 0.255    | 0.681 | 1.303 | 1.684 | 2.021  | 2.423  | 2.704  | 3.307   | 3.551   |  |
| 60       | 0.254    | 0.679 | 1.296 | 1.671 | 2.000  | 2.390  | 2.660  | 3.232   | 3.460   |  |
| 120      | 0.254    | 0.677 | 1.289 | 1.658 | 1.980  | 2.358  | 2.617  | 3.160   | 3.373   |  |
| $\infty$ | 0.253    | 0.674 | 1.282 | 1.645 | 1.960  | 2.326  | 2.576  | 3.090   | 3.291   |  |

Critical value for  
 $\alpha=0.05$ : 2.06

Rejections region:  
Test Statistics < -2.06  
&  
Test Statistics > 2.06

Test statistics:  
-1.9696  
OR -1.92

Distribution of the weights:  
Normal

Sample size ( $n_1, n_2$ ): (12, 16)  
( $s_1, s_2$ ) = (0.5, 0.6)

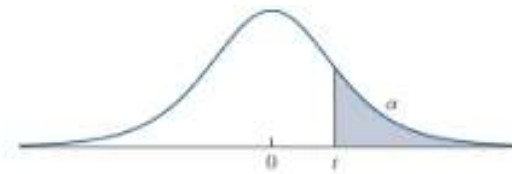
The degrees of freedom is 25.

$$2 \cdot 0.025 < \text{p value} < 2 \cdot 0.05$$

$$0.05 < \text{p value} < 0.10$$

→ Cannot reject  $H_0$  at  $\alpha=5\%$ .

→ Reject  $H_0$  at  $\alpha=10\%$ .



|          |       |       |       | $\alpha$ |        |        |        |         |         |
|----------|-------|-------|-------|----------|--------|--------|--------|---------|---------|
| $v$      | 0.40  | 0.25  | 0.10  | 0.05     | 0.025  | 0.01   | 0.005  | 0.001   | 0.0005  |
| 1        | 0.325 | 1.000 | 3.078 | 6.314    | 12.706 | 31.821 | 63.657 | 318.309 | 636.619 |
| 2        | 0.289 | 0.816 | 1.886 | 2.920    | 4.303  | 6.965  | 9.925  | 22.327  | 31.599  |
| 3        | 0.277 | 0.765 | 1.638 | 2.353    | 3.182  | 4.541  | 5.841  | 10.215  | 12.924  |
| 4        | 0.271 | 0.741 | 1.533 | 2.132    | 2.776  | 3.747  | 4.604  | 7.173   | 8.610   |
| 5        | 0.267 | 0.727 | 1.476 | 2.015    | 2.571  | 3.365  | 4.032  | 5.893   | 6.869   |
| 6        | 0.265 | 0.718 | 1.440 | 1.943    | 2.447  | 3.143  | 3.707  | 5.208   | 5.959   |
| 7        | 0.263 | 0.711 | 1.415 | 1.895    | 2.365  | 2.998  | 3.499  | 4.785   | 5.408   |
| 8        | 0.262 | 0.706 | 1.397 | 1.860    | 2.306  | 2.896  | 3.355  | 4.501   | 5.041   |
| 9        | 0.261 | 0.703 | 1.383 | 1.833    | 2.262  | 2.821  | 3.250  | 4.297   | 4.781   |
| 10       | 0.260 | 0.700 | 1.372 | 1.812    | 2.228  | 2.764  | 3.169  | 4.144   | 4.587   |
| 11       | 0.260 | 0.697 | 1.363 | 1.796    | 2.201  | 2.718  | 3.106  | 4.025   | 4.437   |
| 12       | 0.259 | 0.695 | 1.356 | 1.782    | 2.179  | 2.681  | 3.055  | 3.930   | 4.318   |
| 13       | 0.259 | 0.694 | 1.350 | 1.771    | 2.160  | 2.650  | 3.012  | 3.852   | 4.221   |
| 14       | 0.258 | 0.692 | 1.345 | 1.761    | 2.145  | 2.624  | 2.977  | 3.787   | 4.140   |
| 15       | 0.258 | 0.691 | 1.341 | 1.753    | 2.131  | 2.602  | 2.947  | 3.733   | 4.073   |
| 16       | 0.258 | 0.690 | 1.337 | 1.746    | 2.120  | 2.583  | 2.921  | 3.686   | 4.015   |
| 17       | 0.257 | 0.689 | 1.333 | 1.740    | 2.110  | 2.567  | 2.898  | 3.646   | 3.965   |
| 18       | 0.257 | 0.688 | 1.330 | 1.734    | 2.101  | 2.552  | 2.878  | 3.610   | 3.922   |
| 19       | 0.257 | 0.688 | 1.328 | 1.729    | 2.093  | 2.539  | 2.861  | 3.579   | 3.883   |
| 20       | 0.257 | 0.687 | 1.325 | 1.725    | 2.086  | 2.528  | 2.845  | 3.552   | 3.850   |
| 21       | 0.257 | 0.686 | 1.323 | 1.721    | 2.080  | 2.518  | 2.831  | 3.527   | 3.819   |
| 22       | 0.256 | 0.686 | 1.321 | 1.717    | 2.074  | 2.508  | 2.819  | 3.505   | 3.792   |
| 23       | 0.256 | 0.685 | 1.319 | 1.714    | 2.069  | 2.500  | 2.807  | 3.485   | 3.768   |
| 24       | 0.256 | 0.685 | 1.318 | 1.711    | 2.064  | 2.492  | 2.797  | 3.467   | 3.745   |
| 25       | 0.256 | 0.684 | 1.316 | 1.708    | 2.060  | 2.485  | 2.787  | 3.450   | 3.725   |
| 26       | 0.256 | 0.684 | 1.315 | 1.706    | 2.056  | 2.479  | 2.779  | 3.435   | 3.707   |
| 27       | 0.256 | 0.684 | 1.314 | 1.703    | 2.052  | 2.473  | 2.771  | 3.421   | 3.690   |
| 28       | 0.256 | 0.683 | 1.313 | 1.701    | 2.048  | 2.467  | 2.763  | 3.408   | 3.674   |
| 29       | 0.256 | 0.683 | 1.311 | 1.699    | 2.045  | 2.462  | 2.756  | 3.396   | 3.659   |
| 30       | 0.256 | 0.683 | 1.310 | 1.697    | 2.042  | 2.457  | 2.750  | 3.385   | 3.646   |
| 35       | 0.255 | 0.682 | 1.306 | 1.690    | 2.030  | 2.438  | 2.724  | 3.340   | 3.591   |
| 40       | 0.255 | 0.681 | 1.303 | 1.684    | 2.021  | 2.423  | 2.704  | 3.307   | 3.551   |
| 60       | 0.254 | 0.679 | 1.296 | 1.671    | 2.000  | 2.390  | 2.660  | 3.232   | 3.460   |
| 120      | 0.254 | 0.677 | 1.289 | 1.658    | 1.980  | 2.358  | 2.617  | 3.160   | 3.373   |
| $\infty$ | 0.253 | 0.674 | 1.282 | 1.645    | 1.960  | 2.326  | 2.576  | 3.090   | 3.291   |

# Class Exercise – (4)

---

Distribution of the weights: Unknown

Sample size ( $n_1, n_2$ ): (42, 46)

( $s_1, s_2$ ) = (0.5, 0.6)

$$\text{Test Statistic} = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}} = \frac{5.2 - 5.61}{\sqrt{\frac{0.5^2}{42} + \frac{0.6^2}{46}}} = -3.49$$

Test Stat is approximately normal by the CLT since both sample sizes are greater than 40

The p-value:

$2 * P(Z > |-3.49|) = 2(1 - 0.9998) = 2 * 0.0002 = 0.0004$ ,  
which is smaller than all plausible significance levels

→ Reject  $H_0$  at  $\alpha = 1\%$ .

# Class Exercise – (4)

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Distribution of the weights: Unknown

Sample size ( $n_1, n_2$ ): (42, 46)

( $s_1, s_2$ ) = (0.5, 0.6)

$$\text{Test Statistic} = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}} = \frac{5.2 - 5.6}{\sqrt{\frac{0.5^2}{42} + \frac{0.6^2}{46}}} = -3.419$$

Test Stat is approximately normal by the CLT since both sample sizes are greater than 40

The p-value:

$2 * P(Z > |-3.42|) = 2(1 - 0.9997) = 2 * 0.0003 = 0.0006$ , which is smaller than all plausible significance levels

→ Reject  $H_0$  at  $\alpha = 1\%$ .

[illegible]

# Summary

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| $\bar{X} - \bar{Y} = 5.2 - 5.61$ |              |                  |                  |               |
|----------------------------------|--------------|------------------|------------------|---------------|
|                                  | P value      | $\alpha=1\%$     | $\alpha=5\%$     | $\alpha=10\%$ |
| 1                                | 0.0489       | Cannot reject H0 | Reject H0        | Reject H0     |
| 2                                | ---          |                  |                  |               |
| 3                                | (0.05, 0.10) | Cannot reject H0 | Cannot reject H0 | Reject H0     |
| 4                                | 0.0004       | Reject H0        | Reject H0        | Reject H0     |

| $\bar{X} - \bar{Y} = 5.2 - 5.6$ |              |                  |                  |               |
|---------------------------------|--------------|------------------|------------------|---------------|
|                                 | P value      | $\alpha=1\%$     | $\alpha=5\%$     | $\alpha=10\%$ |
| 1                               | 0.0547       | Cannot reject H0 | Cannot reject H0 | Reject H0     |
| 2                               | ---          | ---              | ---              | ---           |
| 3                               | (0.05, 0.10) | Cannot reject H0 | Cannot reject H0 | Reject H0     |
| 4                               | 0.0004       | Reject H0        | Reject H0        | Reject H0     |

# Inferences concerning a difference between population proportions

$X \sim \text{Bin}(m, p_1)$      $Y \sim \text{Bin}(n, p_2)$  – independent

Estimators for proportions:  $\hat{p}_1 = \frac{X}{m}$  and  $\hat{p}_2 = \frac{Y}{n}$

$$E \left[ \frac{X}{m} - \frac{Y}{n} \right] = p_1 - p_2 \qquad V(\hat{p}_1 - \hat{p}_2) = \frac{p_1 q_1}{m} + \frac{p_2 q_2}{n}$$

large  $m$  &  $n$ : 
$$Z = \frac{\hat{p}_1 - \hat{p}_2 - (p_1 - p_2)}{\sqrt{\frac{p_1 q_1}{m} + \frac{p_2 q_2}{n}}} \sim N(0, 1)$$

# Tests concerning a difference between population proportions

$$H_0: p_1 - p_2 = \Delta_0$$

$$\text{Test statistics: } z_0 = \frac{\hat{p}_1 - \hat{p}_2 - \Delta_0}{\sqrt{\frac{\hat{p}_1 \hat{q}_1}{m} + \frac{\hat{p}_2 \hat{q}_2}{n}}}$$

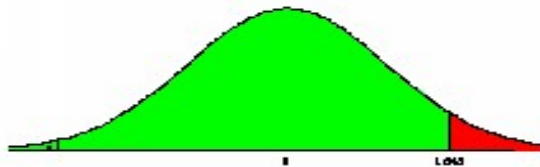
$$H_a: p_1 - p_2 > \Delta_0$$

$$H_a: p_1 - p_2 \neq \Delta_0$$

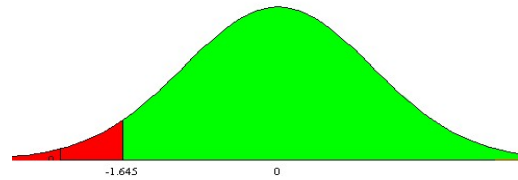
$$H_a: p_1 - p_2 < \Delta_0$$

Accept

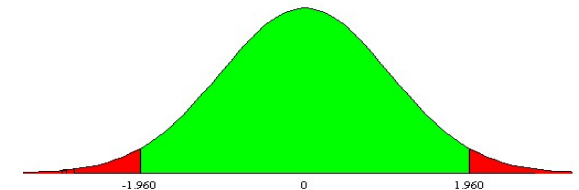
Reject



P-value:  $P(Z \geq z_0)$



P-value:  $P(Z \leq z_0)$



P-value:  $2 \times P(Z \geq |z_0|)$

## Most Common $H_0: p_1=p_2$

- Estimators for proportions:  $\hat{p}_1 = \frac{X}{m}$  and  $\hat{p}_2 = \frac{Y}{n}$
- If  $H_0$  is true,  $p = p_1 = p_2$ , then  $p$  can be estimated by  $\hat{p}$ .

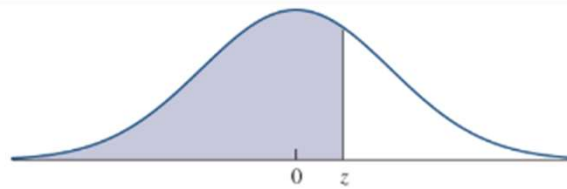
$$\hat{p} = \frac{m}{m+n} \hat{p}_1 + \frac{n}{m+n} \hat{p}_2 = \frac{X+Y}{m+n}$$

- Then, we change our estimator for the standard deviation:

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}\hat{q} \left( \frac{1}{m} + \frac{1}{n} \right)}} \sim N(0,1)$$

# Example

- Depression treatment
- A certain extract versus placebo
- 200 patients: randomly assigned to one of the treatments
- Placebo: 19 patients improved out of 100
- Extract: 27 patients improved out of 100
- State the appropriate hypotheses.
- Find the corresponding p-value.
- What is your conclusion?



| <b>z</b>   | <b>0.00</b> | <b>0.01</b> | <b>0.02</b> | <b>0.03</b> | <b>0.04</b>  | <b>0.05</b> | <b>0.06</b> | <b>0.07</b> | <b>0.08</b> | <b>0.09</b> |
|------------|-------------|-------------|-------------|-------------|--------------|-------------|-------------|-------------|-------------|-------------|
| <b>0.0</b> | .5000       | .5040       | .5080       | .5120       | .5160        | .5199       | .5239       | .5279       | .5319       | .5359       |
| <b>0.1</b> | .5398       | .5438       | .5478       | .5517       | .5557        | .5596       | .5636       | .5675       | .5714       | .5753       |
| <b>0.2</b> | .5793       | .5832       | .5871       | .5910       | .5948        | .5987       | .6026       | .6064       | .6103       | .6141       |
| <b>0.3</b> | .6179       | .6217       | .6255       | .6293       | .6331        | .6368       | .6406       | .6443       | .6480       | .6517       |
| <b>0.4</b> | .6554       | .6591       | .6628       | .6664       | .6700        | .6736       | .6772       | .6808       | .6844       | .6879       |
| <b>0.5</b> | .6915       | .6950       | .6985       | .7019       | .7054        | .7088       | .7123       | .7157       | .7190       | .7224       |
| <b>0.6</b> | .7257       | .7291       | .7324       | .7357       | .7389        | .7422       | .7454       | .7486       | .7517       | .7549       |
| <b>0.7</b> | .7580       | .7611       | .7642       | .7673       | .7704        | .7734       | .7764       | .7794       | .7823       | .7852       |
| <b>0.8</b> | .7881       | .7910       | .7939       | .7967       | .7995        | .8023       | .8051       | .8078       | .8106       | .8133       |
| <b>0.9</b> | .8159       | .8186       | .8212       | .8238       | .8264        | .8289       | .8315       | .8340       | .8365       | .8389       |
| <b>1.0</b> | .8413       | .8438       | .8461       | .8485       | .8508        | .8531       | .8554       | .8577       | .8599       | .8621       |
| <b>1.1</b> | .8643       | .8665       | .8686       | .8708       | .8729        | .8749       | .8770       | .8790       | .8810       | .8830       |
| <b>1.2</b> | .8849       | .8869       | .8888       | .8907       | .8925        | .8944       | .8962       | .8980       | .8997       | .9015       |
| <b>1.3</b> | .9032       | .9049       | .9066       | .9082       | <b>.9099</b> | .9115       | .9131       | .9147       | .9162       | .9177       |
| <b>1.4</b> | .9192       | .9207       | .9222       | .9236       | .9251        | .9265       | .9279       | .9292       | .9306       | .9319       |
| <b>1.5</b> | .9332       | .9345       | .9357       | .9370       | .9382        | .9394       | .9406       | .9418       | .9429       | .9441       |
| <b>1.6</b> | .9452       | .9463       | .9474       | .9484       | .9495        | .9505       | .9515       | .9525       | .9535       | .9545       |
| <b>1.7</b> | .9554       | .9564       | .9573       | .9582       | .9591        | .9599       | .9608       | .9616       | .9625       | .9633       |
| <b>1.8</b> | .9641       | .9649       | .9656       | .9664       | .9671        | .9678       | .9686       | .9693       | .9699       | .9706       |
| <b>1.9</b> | .9713       | .9719       | .9726       | .9732       | .9738        | .9744       | .9750       | .9756       | .9761       | .9767       |
| <b>2.0</b> | .9772       | .9778       | .9783       | .9788       | .9793        | .9798       | .9803       | .9808       | .9812       | .9817       |
| <b>2.1</b> | .9821       | .9826       | .9830       | .9834       | .9838        | .9842       | .9846       | .9850       | .9854       | .9857       |
| <b>2.2</b> | .9861       | .9864       | .9868       | .9871       | .9875        | .9878       | .9881       | .9884       | .9887       | .9890       |
| <b>2.3</b> | .9893       | .9896       | .9898       | .9901       | .9904        | .9906       | .9909       | .9911       | .9913       | .9916       |
| <b>2.4</b> | .9918       | .9920       | .9922       | .9925       | .9927        | .9929       | .9931       | .9932       | .9934       | .9936       |
| <b>2.5</b> | .9938       | .9940       | .9941       | .9943       | .9945        | .9946       | .9948       | .9949       | .9951       | .9952       |
| <b>2.6</b> | .9953       | .9955       | .9956       | .9957       | .9959        | .9960       | .9961       | .9962       | .9963       | .9964       |
| <b>2.7</b> | .9965       | .9966       | .9967       | .9968       | .9969        | .9970       | .9971       | .9972       | .9973       | .9974       |
| <b>2.8</b> | .9974       | .9975       | .9976       | .9977       | .9977        | .9978       | .9979       | .9979       | .9980       | .9981       |
| <b>2.9</b> | .9981       | .9982       | .9982       | .9983       | .9984        | .9984       | .9985       | .9985       | .9986       | .9986       |